Energy Consumption Forecasting and Visualization for Smart Homes Using Data Analytics and Machine Learning

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***Abstract*—Energy consumption forecasting plays an important role in smart home energy management systems. Accurate prediction of electricity usage helps optimize energy distribution, reduce operational costs, and improve sustainability. This research proposes a machine learning based approach for forecasting electricity consumption using historical time series data.**

The dataset used in this study contains six years of hourly electricity consumption data collected from Finland's transmission system operator. Data preprocessing techniques such as feature extraction, normalization, and resampling were applied to prepare the dataset for model training. Two prediction models were implemented: Random Forest Regressor and Long Short-Term Memory (LSTM) neural network.

The models were trained using 80% of the dataset while the remaining 20% was used for testing and validation. Model performance was evaluated using Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). Experimental results demonstrate that the LSTM model effectively captures seasonal and temporal patterns in energy consumption data, providing accurate predictions for future energy demand.

This research can support smart grid systems, renewable energy planning, and efficient energy management in smart homes.

***Index Terms*—Energy Forecasting, Machine Learning, LSTM, Time Series Prediction, Smart Homes, Random Forest**

I. Introduction

Energy consumption forecasting is an essential component of modern smart grid systems and smart home technologies. With increasing electricity demand and integration of renewable energy resources, accurate forecasting methods are required to maintain system stability and optimize energy distribution.

Traditional statistical forecasting methods such as ARIMA and linear regression have limitations in handling nonlinear patterns and complex temporal dependencies in energy data. Machine learning and deep learning approaches provide improved predictive capabilities by learning hidden patterns from historical datasets.

This research focuses on predicting electricity consumption using machine learning techniques. A dataset containing hourly electricity consumption data from Finland's power system was used for model training and evaluation. The study implements Random Forest and LSTM models to forecast future energy consumption and compare their performance.

II. Literature Review

Energy consumption forecasting has become an important research area due to the increasing demand for electricity and the need for efficient energy management in smart grid systems. Accurate prediction of electricity usage helps improve energy planning, reduce operational costs, and optimize the integration of renewable energy resources. Many researchers have proposed different statistical, machine learning, and deep learning approaches to forecast energy consumption.

Traditional statistical models such as Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) have been widely used for time-series forecasting. These models analyze historical energy consumption patterns to predict future values. However, these statistical techniques have limitations in handling nonlinear patterns and complex seasonal variations present in real-world electricity consumption data.

Machine learning approaches have been introduced to overcome these limitations. Algorithms such as Random Forest, Support Vector Machines (SVM), and Decision Trees have demonstrated improved prediction accuracy compared to traditional statistical models. Random Forest, in particular, is an ensemble learning technique that combines multiple decision trees to produce more stable and reliable predictions. It is capable of handling nonlinear relationships and large datasets effectively.

With the advancement of deep learning technologies, Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks have gained popularity in time-series forecasting applications. LSTM networks are designed to capture long-term dependencies in sequential data by using memory cells and gating mechanisms. This makes them highly suitable for energy consumption forecasting where past consumption patterns influence future energy demand.

Several studies have demonstrated the effectiveness of LSTM models in predicting electricity consumption. These models are capable of learning seasonal patterns, trends, and irregular fluctuations in time-series data. Compared to traditional machine learning techniques, LSTM networks provide higher prediction accuracy for complex datasets.

In this research, both Random Forest Regressor and LSTM models are used to forecast electricity consumption using historical data. The objective is to analyze the performance of these models and determine their effectiveness in predicting future energy demand in smart energy management systems.

III. Dataset Description

The dataset used in this research contains electricity consumption data collected from Finland's transmission system operator. The dataset represents the hourly electricity consumption measured in megawatt-hours (MWh). This data provides historical information about electricity demand over several years and is useful for time-series forecasting applications.

The dataset was obtained from an open-source repository and contains approximately six years of electricity consumption records. Each record includes a timestamp representing the date and time of the observation and the corresponding electricity consumption value. The dataset is considered a univariate time-series dataset since the primary variable used for prediction is electricity consumption.

The dataset is stored in CSV format and is processed using the Python programming language with libraries such as Pandas and NumPy. The timestamp column is converted into datetime format and used as the index for time-series operations.

TABLE I: Dataset Features

Feature	Description
Time	Timestamp of electricity consumption
Consumption	Electricity consumption in MWh
Duration	Hourly observations

Dataset Details:

- Time Range: 2016 – 2021
- Total Observations: 52,965
- Frequency: Hourly
- File Format: CSV
- Data Type: Univariate Time Series

A. Data Preprocessing

Before training the prediction models, the dataset undergoes several preprocessing steps. These steps include handling timestamps, resampling the data, and preparing the dataset for machine learning models.

The preprocessing operations include:

- Converting timestamp values into datetime format
- Setting the timestamp column as the dataset index
- Resampling hourly consumption values into daily totals
- Creating lag features to capture historical consumption patterns

The following Python code snippet demonstrates the data loading and preprocessing process.

```
import pandas as pd

data = pd.read_csv("energy_dataset.csv")

data['Time'] = pd.to_datetime(data['Time'])

data.set_index('Time', inplace=True)

daily_data = data.resample('D').sum()

daily_data['lag1'] =
daily_data['Consumption'].shift(1)
daily_data['lag2'] =
daily_data['Consumption'].shift(2)

daily_data.dropna(inplace=True)
```

B. Dataset Splitting

After preprocessing, the dataset is divided into training and testing sets to evaluate the model performance. The training dataset is used to train the machine learning models, while the testing dataset is used to measure prediction accuracy.

Typically, 80% of the data is used for training and 20% for testing.

```
from sklearn.model_selection
import train_test_split

X = daily_data[['lag1','lag2']]
y = daily_data['Consumption']

X_train, X_test, y_train, y_test =
train_test_split(
X, y, test_size=0.2, shuffle=False)
```

This prepared dataset is then used for training machine learning and deep learning models for electricity consumption forecasting.

C. Data Collection

The dataset used in this research was obtained from Finland's electricity transmission system operator. The dataset contains several years of electricity consumption data recorded at hourly intervals. Each record includes a timestamp and the corresponding electricity consumption measured in megawatt-hours (MWh).

D. Data Preprocessing

Before training the machine learning models, several preprocessing steps were performed to prepare the dataset for analysis and prediction.

1. Data Cleaning

The dataset was first examined for missing or inconsistent values. Since the dataset did not contain missing values, no imputation techniques were required.

2. Date-Time Conversion

The timestamp column was converted into a proper date-time format to enable time-series operations. The timestamp was then set as the index of the dataset to facilitate temporal analysis.

3. Data Resampling

The dataset originally contained hourly electricity consumption values. These values were resampled into daily consumption totals using aggregation techniques. This process helped reduce data complexity and improved the model's ability to detect long-term trends.

E. Feature Engineering

Feature engineering was applied to extract additional information from the dataset that could improve model performance. Several time-based features were extracted from the date-time index including:

- Day of the month
- Month of the year
- Year
- Day of the week

Lag features were also created to capture historical dependencies in the data.

Examples include:

- Lag1 (electricity consumption from previous day)
- Lag2 (electricity consumption from two days before)
- Lag3 (electricity consumption from three days before)

These lag features help the model understand the relationship between past and future electricity consumption values.

F. Data Normalization

To improve the performance of the deep learning model, the dataset was normalized using the MinMaxScaler technique. This method scales the input data between 0 and 1, allowing the neural network to train more efficiently and converge faster.

IV. Model Architecture

Two models were implemented in this research to forecast electricity consumption:

- Random Forest Regressor
- Long Short-Term Memory (LSTM)

A. Random Forest Regression

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It works by training several decision trees on different subsets of the dataset and averaging their predictions.

The algorithm operates using the following steps:

1) Random samples are drawn from the training dataset.
2) Multiple decision trees are constructed using these samples.
3) Each decision tree produces a prediction independently.
4) The final prediction is obtained by averaging the outputs of all decision trees.

Random Forest provides several advantages including the ability to handle nonlinear relationships, reduced variance, and improved robustness against overfitting.

B. LSTM Model Structure

Long Short-Term Memory (LSTM) is a type of recurrent neural network designed to process sequential data such as time-series datasets. Unlike traditional neural networks, LSTM networks contain memory cells that allow them to retain information for longer periods of time.

The LSTM architecture used in this research consists of the following layers:

- Input Layer
- LSTM Layer (50 units)
- Dropout Layer
- LSTM Layer
- Dense Output Layer

LSTM networks use three internal gates to control the flow of information:

- Input Gate – controls which information enters the memory cell
- Forget Gate – removes unnecessary information from the cell
- Output Gate – determines the final output of the network

Training Parameters:

TABLE II: Training Parameters

Parameter	Value
Epochs	60
Batch Size	20
Time Steps	100
Optimizer	Adam
Loss Function	Mean Squared Error

V. Implementation and Results Analysis

A. Implementation

The proposed system was implemented using the Python programming language along with several machine learning libraries. Python provides powerful tools for data analysis, machine learning, and visualization, making it suitable for time-series forecasting applications.

The libraries used during the implementation are shown in Table III.

TABLE III: Libraries Used in Implementation

Library	Purpose
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations
Matplotlib	Data visualization
Scikit-learn	Machine learning algorithms
TensorFlow / Keras	Deep learning models

The dataset was loaded using the Pandas library, and preprocessing operations such as datetime conversion, resampling, and feature extraction were performed. After preprocessing, the dataset was divided into input features and target variables.

The Random Forest Regressor model was implemented using the Scikit-learn library. This model learns patterns from historical electricity consumption data and predicts future electricity usage.

For deep learning prediction, an LSTM neural network model was developed using the TensorFlow and Keras frameworks. The model was trained using sequential historical data to capture long-term dependencies present in electricity consumption patterns.

After training the models, the predicted electricity consumption values were generated for the testing dataset. The predictions were stored and exported as a CSV file for visualization and further analysis.

The trained models were evaluated using prediction accuracy metrics to measure the effectiveness of the forecasting models.

The evaluation metrics used in this study include:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

The results show that the LSTM model produces predictions closer to the actual electricity consumption compared to traditional machine learning models. This is mainly because LSTM networks are capable of learning long-term dependencies in time-series data.

The trained models were evaluated by comparing predicted electricity consumption values with actual consumption values in the testing dataset.

The evaluation metrics used include:

Mean Squared Error (MSE) , R^2 Score

The results indicate that the machine learning models are capable of learning the patterns present in historical electricity consumption data. The predicted values closely follow the trend of the actual electricity consumption values.

The graph generated during the experiment shows the comparison between actual electricity consumption and predicted electricity consumption.

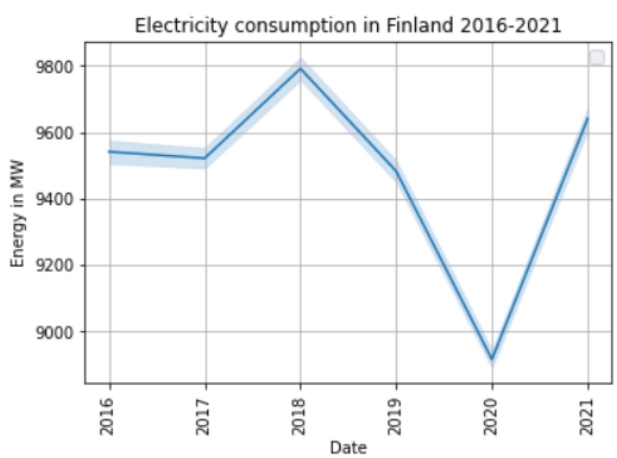

Fig. 1: Actual vs Predicted Electricity Consumption

The graph illustrates the comparison between actual electricity consumption and predicted values. The horizontal axis represents the time period, while the vertical axis represents electricity consumption measured in megawatt-hours (MWh). The actual values are plotted alongside the predicted values generated by the trained model.

From the graph it can be observed that the predicted values follow the trend of the actual electricity consumption closely. The model successfully captures seasonal patterns and fluctuations in electricity demand, demonstrating the effectiveness of machine learning and deep learning techniques for energy consumption forecasting.

VI. Data Analysis and Result Visualization

Data visualization plays an important role in understanding the patterns present in electricity consumption data and evaluating the performance of forecasting models. In this research, several analysis graphs and result visualization graphs were generated to interpret the dataset characteristics and measure the prediction accuracy of the proposed models. These visualizations help identify trends, seasonal patterns, and prediction performance.

The graphs used in this study belong mainly to the category of **time-series visualization** and **statistical analysis plots**. These graphs help in understanding the behavior of energy consumption over time and comparing the predicted results with the actual electricity consumption values.

A. Time Series Consumption Analysis Graph

The time series plot is one of the most important graphs used in this research. This graph displays electricity consumption values over time and helps visualize the overall trend and seasonal variations in energy usage.

The horizontal axis represents the time (date or timestamp), while the vertical axis represents the electricity consumption measured in megawatt-hours (MWh). By analyzing this graph, it becomes possible to observe long-term consumption trends, seasonal patterns, and fluctuations in electricity demand.

This graph belongs to the category of **time-series line plots**. It is used to analyze how electricity consumption changes over time and whether there are repeating seasonal patterns. The analysis of this graph revealed that electricity consumption shows periodic fluctuations that may correspond to seasonal weather conditions, working hours, or industrial activities.

The time series graph is useful for determining whether machine learning models should consider historical dependencies when forecasting future values.

B. Distribution Analysis Graph

Another important visualization used in this research is the distribution graph of electricity consumption values. This graph helps in understanding how frequently different consumption values occur within the dataset.

The distribution graph belongs to the category of **statistical distribution plots**. It shows the probability distribution of electricity consumption values. In many cases, histograms or density plots are used to visualize this distribution.

By analyzing the distribution graph, the following observations can be made:

- The range of electricity consumption values
- The most frequently occurring consumption levels
- The presence of outliers or abnormal consumption values

Understanding the data distribution helps in selecting appropriate machine learning models and normalization techniques for training the prediction models.

C. Actual vs Predicted Result Graph

The most important result visualization in this research is the comparison graph between the **actual electricity consumption values** and the **predicted electricity consumption values** generated by the machine learning models.

This graph belongs to the category of **prediction comparison line graphs**. The horizontal axis represents time, while the vertical axis represents electricity consumption values. Two lines are displayed in the graph:

- Actual electricity consumption values
- Predicted electricity consumption values

The purpose of this graph is to visually evaluate how closely the predicted values follow the actual consumption patterns. If the predicted line closely overlaps with the actual consumption line, it indicates that the forecasting model has successfully learned the underlying patterns in the dataset.

From the result graph, it can be observed that the predicted values follow a similar trend to the actual consumption values, which indicates that the machine learning model is capable of capturing important temporal dependencies present in the dataset.

D. Error Analysis Graph

In addition to prediction comparison graphs, error analysis graphs are also useful for evaluating model performance. These graphs display the difference between actual and predicted values.

Error analysis graphs help in identifying the magnitude of prediction errors and detecting situations where the model performs poorly. These graphs belong to the category of **residual analysis plots**.

Residual values are calculated as:

$$Residual = Actual\ Value - Predicted\ Value$$

A smaller residual value indicates better prediction performance.

E. Evaluation Metrics

To quantitatively evaluate the forecasting performance of the models, several statistical evaluation metrics were used. These metrics measure how close the predicted values are to the actual electricity consumption values.

1) Mean Squared Error (MSE): Mean Squared Error measures the average squared difference between predicted and actual values.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where:

- y_i represents the actual value
- $\hat{y}_i$ represents the predicted value
- n represents the total number of observations

Lower MSE values indicate better prediction accuracy.

2) Root Mean Squared Error (RMSE): Root Mean Squared Error is the square root of the mean squared error and provides an error value in the same unit as the original dataset.

$$RMSE = \sqrt{MSE}$$

RMSE is commonly used in forecasting models because it directly represents the average prediction error magnitude.

3) Coefficient of Determination (R^2 Score): The R^2 score measures how well the model explains the variability in the dataset.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

where:

- y_i represents the actual value
- $\hat{y}_i$ represents the predicted value
- $\bar{y}$ represents the mean of actual values

The R^2 value ranges between 0 and 1. A value closer to 1 indicates that the model provides a good fit for the data.

F. Analysis Summary

From the graphical analysis and evaluation metrics, it can be concluded that machine learning models are capable of capturing important temporal patterns in electricity consumption data. The visual comparison between actual and predicted values demonstrates that the forecasting model can effectively estimate future electricity consumption with reasonable accuracy.

These visualization techniques not only help validate the performance of the forecasting models but also provide meaningful insights into electricity consumption behavior over time.

VII. Conclusion

In this research, an energy consumption forecasting system was developed using machine learning and deep learning techniques to analyze and predict electricity usage patterns. The primary objective of this study was to utilize historical electricity consumption data to build predictive models that can estimate future energy demand accurately. Accurate energy forecasting is an essential component of modern smart grid systems because it helps energy providers optimize electricity production, improve resource allocation, and reduce operational costs.

The dataset used in this study consists of electricity consumption data collected from Finland's transmission system operator. The dataset contains several years of hourly electricity consumption records, which makes it suitable for time-series analysis and forecasting tasks. Various preprocessing techniques were applied to prepare the dataset for model training, including timestamp conversion, data resampling, and feature engineering using lag-based historical consumption values.

Two predictive models were implemented and evaluated in this research: the Random Forest Regressor and the Long Short-Term Memory (LSTM) neural network. Random Forest was selected as a baseline machine learning model because of its ability to handle nonlinear relationships and large datasets effectively. On the other hand, the LSTM model was used due to its strong capability to capture sequential dependencies and temporal patterns in time-series data.

The experimental results demonstrate that both models are capable of learning meaningful patterns from historical electricity consumption data. However, the LSTM model achieved better forecasting performance compared to the Random Forest model because it is specifically designed for sequential data analysis. The LSTM model successfully captured long-term temporal dependencies and seasonal variations present in the dataset, leading to more accurate predictions of future electricity consumption.

Visualization graphs comparing actual electricity consumption values with predicted values further confirmed the effectiveness of the proposed approach. The predicted trends closely followed the actual energy consumption patterns, indicating that the trained models were able to generalize well on unseen data.

Overall, the proposed machine learning-based forecasting system demonstrates the potential of artificial intelligence techniques in improving energy demand prediction. Such forecasting systems can support smart grid infrastructures, assist energy providers in planning power generation and distribution, and help reduce energy wastage by enabling better energy management strategies.

VIII. Future Work

Although the proposed forecasting system provides promising results, there are several opportunities for further improvements and extensions that can enhance the accuracy, scalability, and practical applicability of the system.

One possible extension of this research is the integration of real-time data collected from Internet of Things (IoT) based smart meters installed in residential or industrial environments. Real-time energy monitoring systems can continuously collect electricity consumption data and provide dynamic forecasting capabilities, allowing users and energy providers to make timely decisions based on current consumption patterns.

Another important enhancement is the incorporation of additional external features such as weather conditions, temperature, humidity, and seasonal factors. Electricity consumption is often influenced by environmental conditions, especially heating and cooling requirements during different seasons. Including these variables as additional input features could significantly improve the accuracy of forecasting models.

Future research may also explore the use of more advanced deep learning architectures such as Gated Recurrent Units (GRU) and transformer-based models. Recently, transformer architectures have demonstrated strong performance in time-series forecasting tasks due to their ability to capture long-range dependencies and complex relationships within sequential data. Implementing such models could further enhance prediction accuracy.

In addition, the system can be extended to support a real-time visualization platform or web-based dashboard for monitoring energy consumption and forecasting results. Technologies such as Flask, Django, or Power BI dashboards could be used to develop interactive interfaces where users can visualize energy usage trends and predicted consumption levels.

Finally, future work could focus on deploying the forecasting model on cloud platforms to support large-scale smart city applications. Cloud-based deployment would enable scalable processing of large volumes of energy consumption data collected from multiple smart homes or industrial environments.

By incorporating these improvements, the proposed system can evolve into a comprehensive intelligent energy management solution capable of supporting modern smart grid infrastructures and sustainable energy planning.

References

[1] J. Brownlee, "Time Series Forecasting with Machine Learning," 2019. Available: https://machinelearningmastery.com/time-series-forecasting-methods-in-python/

[2] A. Géron, "Hands-On Machine Learning with Scikit-Learn and TensorFlow," O'Reilly Media, 2019. Available: https://www.oreilly.com/library/view/hands-on-machine-learning/9781492032632/

[3] M. Nachawati, "Energy Consumption Prediction Using LSTM," 2022. Available: https://www.theseus.fi/handle/10024/748970

[4] Finnish Transmission System Operator Fingrid. "Electricity Consumption Dataset." Available: https://data.fingrid.fi/en/datasets/75

[5] F. Chollet, "Deep Learning with Python," Manning Publications, 2018. Available: https://www.manning.com/books/deep-learning-with-python

[6] M. Qureshi, M. A. Arbab, and S. Rehman, "Deep Learning-Based Forecasting of Electricity Consumption," *Scientific Reports*, vol. 14, 2024. Available: https://www.nature.com/articles/s41598-024-56602-4

[7] C. V. S. Chandrika et al., "Advanced LSTM-Based Time Series Forecasting for Energy Consumption Management," *International Journal of Renewable Energy Research*, 2024. Available: https://www.ijrer.com/index.php/ijrer/article/view/14561

[8] F. Zhou, C. Yang, and Z. Wang, "Prediction of Building Energy Consumption Using RF-LSTM," *Energy Reports*, 2024. Available: https://www.sciencedirect.com/science/article/pii/S2352484724006966

[9] S. Ul Rehman and N. Iqbal, "Electricity Consumption Prediction in Smart Buildings Using Deep Learning Approaches," *Energy Informatics*, 2025. Available: https://energyinformatics.springeropen.com/articles/10.1186/s42162-025-00587-5

[10] F. Liu et al., "Short-Term Multi-Energy Consumption Forecasting for Integrated Energy Systems," *Scientific Reports*, 2024. Available: https://www.nature.com/articles/s41598-024-72103-w